

Traffic Data: Insights Report

Finding 1: Traffic follows the working day

The clearest pattern in the whole dataset is the daily rhythm. Traffic is very light overnight, climbs quickly through the morning, stays busy through the middle of the day, and peaks again in the late afternoon before dropping off at night. The busiest hour has about 15 times more traffic than the quietest hour.

Time of day	Average traffic (2017)
Middle of the night (about 3 a.m.)	382
Morning rush (about 8 a.m.)	4,880
Middle of the day	around 4,400
Evening rush (about 4 p.m.)	5,798

One detail is worth noticing: **the evening rush is busier than the morning rush**. Traffic at 4 p.m. (5,798) is higher than at 8 a.m. (4,880), and it stays heavy from about 3 p.m. to 6 p.m. So the late afternoon is when the roads are under the most pressure.

Finding 2: Weather makes a small difference

Traffic does change a little with the weather. It is highest when it is cloudy (3,618 on average) and lowest during a squall (2,062). That is a gap of about 1,557. It sounds like a lot, but it is smaller than the change we see across a single afternoon. In other words, the shift from a quiet hour to a busy hour matters much more than the shift from clear skies to bad weather.

Finding 3: Temperature tells us almost nothing

When we plot traffic against temperature, the points form a wide blur with no real pattern. There is a very faint tendency for busier roads on warmer days, but it is so weak it is not useful. The likely reason is simple: warm hours and busy hours both happen in the daytime and in summer, so they line up by coincidence. Warmer weather does not actually put more cars on the road.

Finding 4: Small changes across months and years

- **Traffic grew slightly over time.** 2017 was generally the busiest of the three years.
- **Every July is quieter.** Traffic dips in mid-summer each year, most noticeably in 2016. This is probably due to holidays and should be checked against the calendar.
- **Month-to-month changes are small.** The differences between months are minor compared with the difference between morning and evening. The daily pattern is the one that counts.

What this means for the mobility team

- **Plan around the two rush hours, and give extra attention to the evening.** Traffic signals, lane use, and public transport should be set up for the 7–9 a.m. and 3–6 p.m. windows, with more capacity kept for the afternoon.
- **Leave temperature out of traffic forecasts.** It does not help predict traffic. Time of day, day of the week, and season are far more useful.
- **Use the quiet times for roadworks.** Late night (midnight to 5 a.m.) and the July lull are the best times for repairs, closures, and reduced service.

In short: the time of day explains traffic far better than the weather or temperature. Focusing effort on the morning and evening rush, especially the evening, will do more to ease congestion than anything based on the weather.